Sustainability aware asset management

Lecture 4: Regularization of Covariance Matrix and Adding Constraints

Eric Jondeau

Faculté des HEC, Université de Lausanne
Master of Science in Finance

Objectives of the Lecture

Solving the mean-variance optimization requires the estimation of some inputs:

- the risk premia (excess returns forecasts)
- the covariance matrix (or variances and correlations)

One of the main issues with this approach is that portfolios are highly concentrated in a small number of assets, contradicting the common-sense notion of diversification.

Reasons for this result are:

- **Input sensitivity:** Model's output (weights) changes significantly due to small changes in inputs.
- **Estimation error:** Moments of return distribution are estimated from a finite sample, and as such, with uncertainty.
- **Model error:** The mean and the covariance matrix are estimated from past data under some assumptions (iid, normality, etc.).

Objectives of the Lecture

In this lecture, we analyze several approaches to deal with estimation error:

- Constrained portfolio optimization (practitioner-based heuristics)
- Portfolio resampling
- Factor models

- Shrinkage (decision theory)
- Bayesian techniques

- Robust optimization (portfolio construction process)

Readings

- **Michaud and Michaud** (2008), *Estimation Error and Portfolio Optimization*, Journal of Investment Management, First Quarter. (Description of portfolio resampling)

- Three papers by **Ledoit and Wolf** (Description of shrinkage):
  - *Improved Estimation of the Covariance Matrix of Stock Returns with an Application to Portfolio Selection*. *Journal of Empirical Finance*, 10 (2003), 603–621.
  - *Honey, I Shrunk the Sample Covariance Matrix*. *Journal of Portfolio Management*, (2004), 110–119.
  - *A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices*. *Journal of Multivariate Analysis*, 88 (2004), 365–411.

Objectives of the Lecture

- **Constrained Portfolio Optimization**
- Portfolio Resampling
- Factor Models
- Stein's Paradox
- Shrinkage Estimator

Constrained Portfolio Optimization

The most common procedure for avoiding unreasonable solutions is the **constrained portfolio optimization**.

Constraining the set of possible solutions will lead to a shift of the efficient frontier downwards and to the right. However, in practice, portfolio constraints truncate the extreme portfolio weights and thereby improve the performance of the allocations.

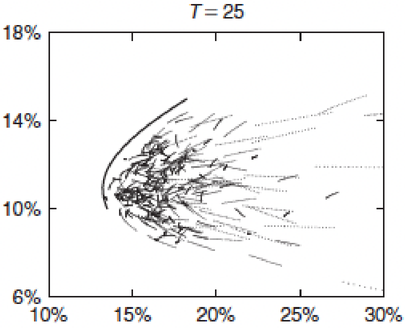
Constraints can be motivated by legal constraints or preferences, but also to safeguard against unreasonable solutions caused by estimation error in inputs. They can be used to enforce some diversification.

Typical constraints:

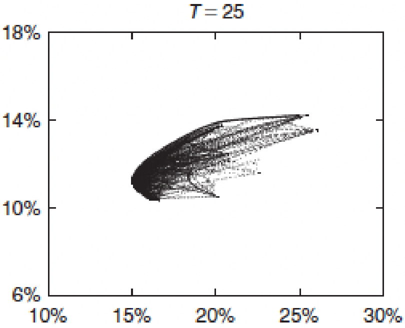
- no short sales
- maximum stock/industry/classes exposure (e.g., $\alpha_i \leq \bar{\alpha}$, say 0.5)
- discrete investment levels (e.g., $\alpha_i \in \{0, 1\}$ or $\alpha_i \in \{0; 25\%; 50\%; 75\%\}$)
- restriction on turnover
- transaction cost management

Constrained Portfolio Optimization – Effect of short-sell constraint

Unconstrained



Constrained



Source: Jobson and Korkie (1980)

Screening

Basic screening procedures:

- rank stocks on alpha
- choose first k stocks
- equal weight (or capitalization weight) for these stocks

Some refinements:

- buy, hold, and sell categories
- different weights for different screen levels (buy vs strong buy)
- apply within sector or style categories $\rightarrow$ stratification

Advantages:

- simple, intuitive, easily customized
- robust to estimation / optimization error
- reduce turnover and transaction costs

Disadvantages:

- ignore incremental information in alpha
- lack integrated risk control
- relatively ad-hoc

Screening: Examples

1- Exclusion based on carbon emissions (see Lecture 7)

- Step 1: sort all firms by increasing carbon intensity
- Step 2: exclude all firms with a carbon intensity above a given threshold (say 95%)
- Step 3: reallocate the proceeds (proportionate or in firms with lowest carbon intensity)

2- Turnover management

Turnover constraint can be specified as: $TO_t = \sum_{i=1}^{N} \left| \alpha_{i,t}^{\text{new}} - \alpha_{i,t}^{\text{old}} \right| \leq \overline{\text{TO}}$

where $\alpha_{i,t}^{\text{old}}$ is the weight of asset i just before the rebalancing.

3- Transaction cost management

Expected return is changed to: $\mu_{i,t}^{\text{net}} = \mu_{i,t} - \, c_i^- \, \Delta\alpha_{i,t}^- - \, c_i^+ \, \Delta\alpha_{i,t}^+$

where $\Delta\alpha_{i,t}^- = \max(0, -\Delta\alpha_{i,t})$ indicates a negative weight change, $\Delta\alpha_{i,t}^+ = \max(0, \Delta\alpha_{i,t})$ indicates a positive weight change, and c_i^- and c_i^+ are the bid and ask transaction costs.

Objectives of the Lecture

- Constrained Portfolio Optimization
- **Portfolio Resampling**
- Factor Models
- Stein's Paradox
- Shrinkage Estimator

Portfolio Resampling

Estimated parameters $\widehat{\boldsymbol{\mu}}$ and $\widehat{\boldsymbol{\Sigma}}$ are calculated using only one possible realization of a return history. When the sample is small, the difference between the point estimates and the expected value can be large.

An algorithm that takes point estimates as inputs and treats them as if they were known will react to tiny differences in returns that are well within measurement error.

The effect of estimation error on optimal portfolio can be captured by a Monte-Carlo procedure called **Resampling**.

Resampling has been introduced by Michaud (1998) and is the subject of U.S. Patent.

The goal of resampling is the reduction of efficient portfolios sensitivity to data. It also allows to clearly distinguish the impact of the uncertainty due to estimation errors in means from that due to estimation error in variance.

Main idea: Construct an efficient frontier by averaging the portfolio weights of mean-variance efficient portfolios in the resampled data.

Portfolio Resampling – Practical receipt

Step 1. Take a sample P_0 of historic returns. Estimate from this sample mean returns $\boldsymbol{\mu}_0$ and covariance matrix $\boldsymbol{\Sigma}_0$.

Estimate the mean-variance efficient frontier for this data, varying the risk aversion λ:

Optimal portfolio weights are $\boldsymbol{\alpha}_0^*(\lambda) = (1/\lambda)\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0$ and portfolio return and variance are $\boldsymbol{\mu}_{0,p} = \boldsymbol{\alpha}_0^*(\lambda)'\boldsymbol{\mu}_0$ and $\sigma_{0,p}^2 = \boldsymbol{\alpha}_0^*(\lambda)'\boldsymbol{\Sigma}_0\boldsymbol{\alpha}_0^*(\lambda)$

Step 2. Assume that the sample P_0 is the realization of a set of iid random variables. Assume that the distribution of these variables is $\mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$

From this distribution, generate a new sample P_1 (with same length as P_0), estimate its mean returns $\boldsymbol{\mu}_1$ and covariance matrix $\boldsymbol{\Sigma}_1$ and solve the optimization problem.

Optimal portfolio weights are $\boldsymbol{\alpha}_1^*(\lambda) = (1/\lambda)\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\mu}_1$ and portfolio return and variance are $\mu_{1,p} = \boldsymbol{\alpha}_1^*(\lambda)'\boldsymbol{\mu}_1$ and $\sigma_{1,p}^2 = \boldsymbol{\alpha}_1^*(\lambda)'\boldsymbol{\Sigma}_1\boldsymbol{\alpha}_1^*(\lambda)$, conditional on the sample

Portfolio Resampling – Practical receipt

Step 3. On this efficient frontier, choose m efficient portfolios with equally-spaced target expected returns $(\tilde{\mu}_i,\ i = 1, \cdots, m)$ starting from minimal-variance portfolio to maximal-return portfolio.

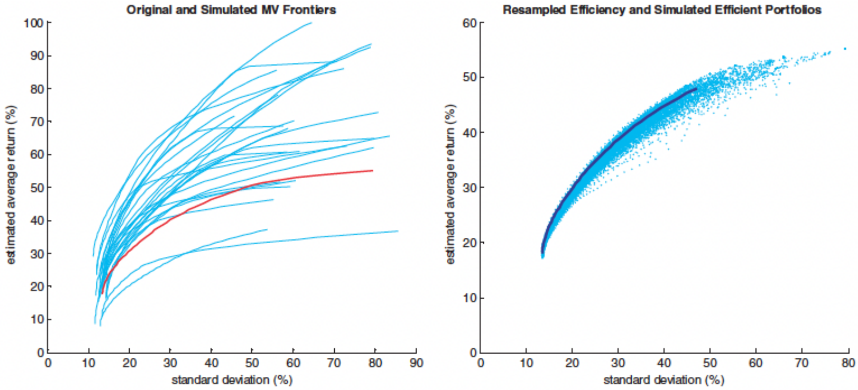
For each of these m portfolios, find asset's weights $\boldsymbol{\alpha}_1(\tilde{\mu}_i),\ i = 1, \cdots, m$, where index 1 indicates that $\boldsymbol{\alpha}_1$ are weights for sample P_1. The corresponding efficient frontier is given by portfolio return $\tilde{\mu}_{1,p} = \boldsymbol{\alpha}_1(\tilde{\mu})'\boldsymbol{\mu}_0$ and variance $\tilde{\sigma}^2_{1,p} = \boldsymbol{\alpha}_1(\tilde{\mu})'\boldsymbol{\Sigma}_0\boldsymbol{\alpha}_1(\tilde{\mu})'$.

Step 4. Generate a new sample and repeat steps 2 to 3 for a large number Q of Monte Carlo simulations (Q should be of order 100–1000).

Step 5. Calculate resampled weights: $\boldsymbol{\alpha}_{res}(\tilde{\mu}_i) = \frac{1}{Q}\sum_{q=1}^{Q}\boldsymbol{\alpha}_q(\tilde{\mu}_i),\ i = 1, \cdots, m$.

Recover the resampled efficient frontier from weights $\boldsymbol{\alpha}_{res}(\tilde{\mu})$ and the sample mean returns $\boldsymbol{\mu}_0$ and covariance matrix $\boldsymbol{\Sigma}_0$: It is defined by portfolio return $\mu_{res,p} = \boldsymbol{\alpha}_{res}(\tilde{\mu})'\boldsymbol{\mu}_0$ and variance $\sigma^2_{res,p} = \boldsymbol{\alpha}_{res}(\tilde{\mu})'\boldsymbol{\Sigma}_0\boldsymbol{\alpha}_{res}(\tilde{\mu})$

Michaud and Michaud (2008), *Estimation Error and Portfolio Optimization*, Journal of Investment Management



Red: Initial efficient frontier (step 1)
Blue: 25 simulated efficient frontier (step 2)

Blue: $m = 51$ efficient portfolios (step 3)
Black: Resampled efficient frontier (step 5)

Illustration – Classical vs resampled efficient frontier

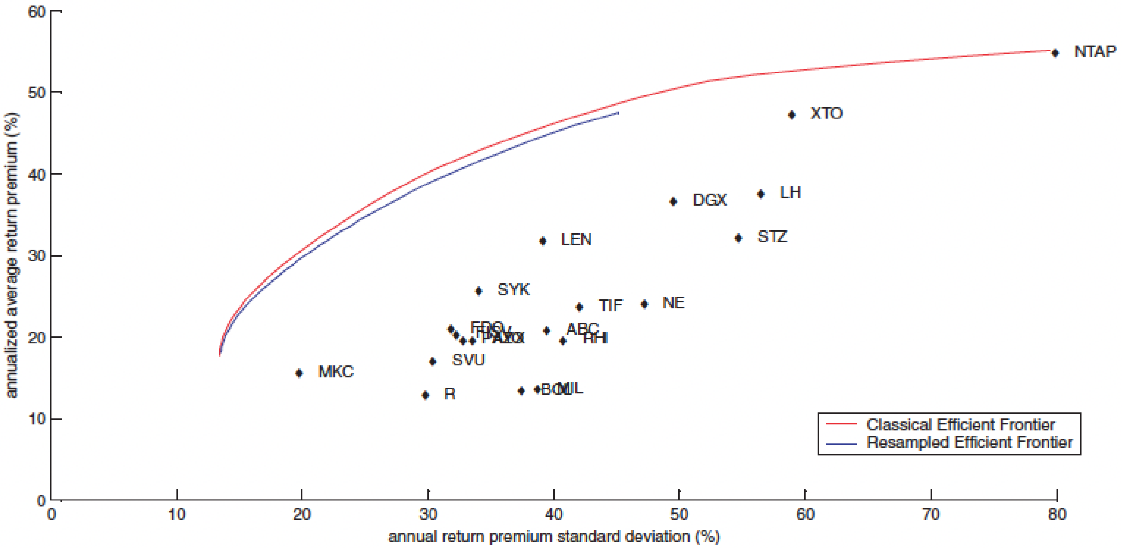
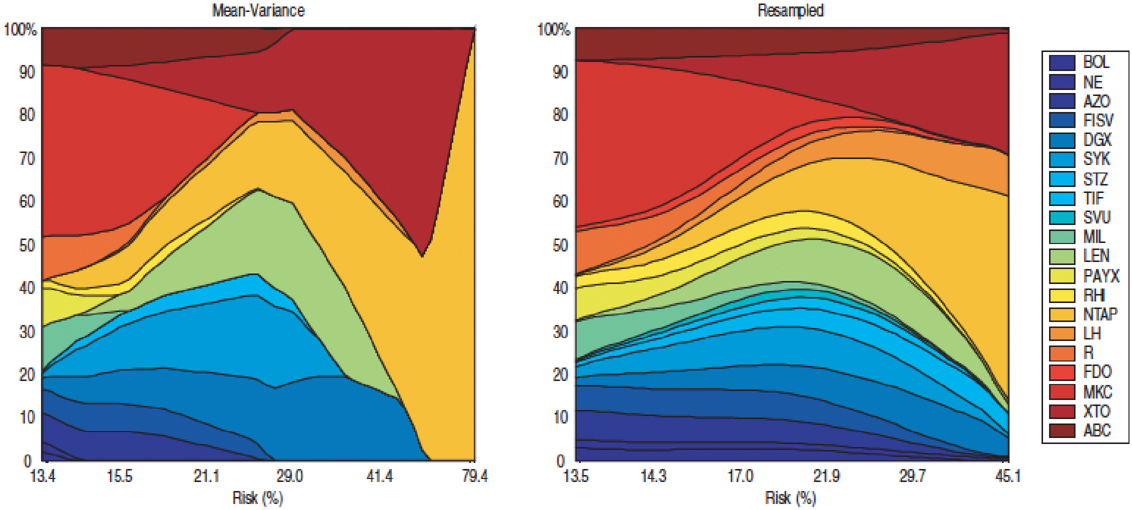


Illustration – Classical and Resampled Portfolio Compositions



Portfolio Resampling – Comments

- Resampled efficient frontier is an average of the weights of the portfolios of the efficient frontiers, which in turn were generated from input obtained from the simulation processes.
- Resampled efficient frontier lies below the true efficient frontier.
- Resampled portfolios show a higher diversification, with more assets entering the solution. This is unlike Markowitz, in which the process of optimization leads to the identification of a particular dominant asset compared with others.
- They also exhibit less sudden shifts in allocations, giving a smooth transition as return requirements change.
- With portfolio constraints:
    - solution is more non-linear
    - resampling is less effective

Objectives of the Lecture

- Constrained Portfolio Optimization
- Portfolio Resampling
- **Factor Models**
- Stein's Paradox
- Shrinkage Estimator

Factor Models

In Lecture 2, we have discussed the factor zoo, which is used to determine expected returns in the cross-section.

A factor model is also a way to reduce the estimation error because it reduces the dimension of the estimation problem. Assume the single-factor model:

$$R_{i,t+1} = \alpha_i + \beta_i f_{t+1} + \varepsilon_{i,t+1},$$

with $\varepsilon_{i,t+1}$ i.i.d. and uncorrelated across stocks and with factor f_{t+1}.

The covariance matrix of returns is:

$$\mathbf{\Sigma} = \mathrm{Var}[\mathbf{R}_{t+1}] = \boldsymbol{\beta}\boldsymbol{\beta}' \, \sigma_f^2 + \mathbf{D}$$

where $\mathbf{D}$ is the diagonal matrix with elements $\sigma_i^2 = \mathrm{Var}[\varepsilon_{i,t+1}]$.

It reduces the estimation problem to $3n+1$ terms:

- estimate α_i and β_i by regressing $R_{i,t+1}$ on f_{t+1}
- estimate σ_f^2 and σ_i^2 with sample variances.

Market Models – Model and Assumptions

In the market model, the return on a stock is written as:

$$R_{i,t}+1 = \alpha_i + \beta_i R_{m,t+1} + \varepsilon_{i,t+1}.$$

First assumption: $\mathrm{Cov}[\varepsilon_{i,t}, R_{m,t}] = 0$

- Implication: how well market model describes the return on stock i is independent on what the market does.
- If parameters are estimated from time-series regressions, residuals $\hat{\varepsilon}_{i,t}$ and $R_{m,t}$ are uncorrelated by construction.

Second assumption: $\mathrm{Cov}(\varepsilon_{i,t}, \varepsilon_{j,t}) = 0, \qquad i \neq j$

- Implication: the only reason why stocks move together is because of their common co-movements with the market.
- This is a strong assumption. The quality of the model will highly depend on whether it is a good approximation of reality or not.

Market Models – Implications

The expected return has two components: $\mathbb{E}[R_i] = \alpha_i + \beta_i\,\mathbb{E}[R_m]$

The variance has TWO components: $\mathrm{Var}(R_i) = \beta_i^2\,\mathrm{Var}(R_m) + \mathrm{Var}(\varepsilon_i)$

The covariance has ONE component: $\mathrm{Cov}(R_i, R_j) = \beta_i \beta_j\,\mathrm{Var}(R_m)$

Covariance matrix:

$$\boldsymbol{\Sigma} = \boldsymbol{\beta}\boldsymbol{\beta}'\,\sigma_m^2 + \mathbf{D}, \qquad \text{where} \qquad \mathbf{D} = \begin{pmatrix} \sigma_{\varepsilon,1}^2 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_{\varepsilon,N}^2 \end{pmatrix}$$

It reduces the estimation problem to $3n + 1$ terms:

- estimate α_i and β_i by regressing $R_{i,t}$ on $R_{m,t}$
- estimate σ_m^2 and $\sigma_{\varepsilon,i}^2$ with sample variances.

Multi-Factor Models

Assume now a model with K factors: $R_{i,t+1} = \alpha_i + \beta_i' \mathbf{f}_{t+1} + \varepsilon_{i,t+1}$

with $\varepsilon_{i,t+1}$ i.i.d. and uncorrelated across stocks and with factors f_{t+1}.

The covariance matrix of returns is: $\boldsymbol{\Sigma} = \mathrm{Var}[\mathbf{R}_{t+1}] = \mathbf{B}\,\boldsymbol{\Sigma}_f\,\mathbf{B}' + \mathbf{D}$

where B is the $N \times K$ matrix of factor loadings, $\boldsymbol{\Sigma}_f$ is the $K \times K$ factor covariance matrix, and D is diagonal (idiosyncratic variances).

It reduces the estimation problem to:

- $K(K+1)/2 + (K+2)N$ terms if factors are correlated
- $K + (K+2)N$ terms if factors are uncorrelated.

Remark: With 500 assets and 600 months of data, with five factors, there are 3515 coefficients to estimate if factors are correlated, as opposed to 125,250 without factors. This translates into a more than 35-fold increase in the degrees of freedom (from less than 2.4 to more than 85).

Multi-Factor Models – Empirical evidence

Chan, Karceski, Lakonishok (1999), *On Portfolio Optimization: Forecasting Covariances and Choosing the Risk Model*, RFS 12(5), 937–974.

Experiment: each year, from 1973 to 1997, estimate covariance matrix for 250 random stocks from past 60 month returns. Form global minimum variance portfolio with short-sell constraints. One-year out-of-sample performance.

	Mean (annual)	Std dev. (annual)	Sharpe ratio	Correlation with market	Nb stocks with weight above 0.5%
Full covariance	15.5	12.9	0.64	0.88	53
1 factor	16.1	12.9	0.69	0.80	54
3 factors	15.7	12.7	0.67	0.82	54
9 factors	15.3	12.9	0.62	0.86	54
Industry+Size	16.1	12.8	0.69	0.83	61
Value weighted	14.3	15.5	0.45	0.98	45
Equal weighted	17.3	16.6	0.60	0.92	0

Multi-Factor Models – Comments

- Factor models perform as well as the plug-in (full covariance) estimates.
- But the various factor models generally provide similar results, both in terms of the number and the choice of factors.
- A simple CAPM-based single-factor model performs only marginally worse than a high-dimensional model with industry and characteristic-based factors.
- The dominant factor (the market portfolio) tends to dominate the remaining factors, so that the incremental informativeness becomes very difficult to detect.

Objectives of the Lecture

- Constrained Portfolio Optimization
- Portfolio Resampling
- Factor Models
- **Stein's Paradox**
- Shrinkage Estimator

Stein's Paradox

We consider the estimation of the mean $\boldsymbol{\mu}$ of a vector of N multivariate normal random variables X_i distributed i.i.d. $X_i \sim \mathcal{N}(\mu_i, 1)$. The difficulty is that we use only one draw.

- If $N = 1$, we draw X_1 from $\mathcal{N}(\mu_1, 1)$. The best estimator is $\widehat{\boldsymbol{\mu}} = \widehat{\mu}_1 = X_1$ (obvious)
- If $N = 2$, we draw $(X_1, X_2)'$ from $\mathcal{N}((\mu_1, \mu_2)', I_2)$. The best estimator is $\widehat{\boldsymbol{\mu}} = (\widehat{\mu}_1, \widehat{\mu}_2)' = (X_1, X_2)'$.
- If $N = 3$, we have a draw $(X_1, X_2, X_3)'$ from $\mathcal{N}((\mu_1, \mu_2, \mu_3)', I_3)$. The best estimator is *NOT* $(X_1, X_2, X_3)'$.

Properties:

- $\widehat{\boldsymbol{\mu}}$ is optimal for $N = 1$ and $N = 2$. No other estimator $\bar{\boldsymbol{\mu}}$ has lower risk for all possible values of $\boldsymbol{\mu}$
- $\widehat{\boldsymbol{\mu}}$ is not optimal for $N \geq 3$. Some estimators $\bar{\boldsymbol{\mu}}$ have lower risk for all possible values of $\boldsymbol{\mu}$

Stein's paradox (1956)

Combining pieces of information a priori independent from each other can help predict individual moments.

James-Stein shrinkage estimators

Loss function to capture statistician's disutility from estimation error. A typical loss function is squared error loss:

$$L(\widehat{\boldsymbol{\mu}}, \boldsymbol{\mu}) = \sum_{i=1}^{N} (\mu_i - \widehat{\mu}_i)^2.$$

Risk function: expected loss as function of the true parameters:

$$R(\boldsymbol{\mu}) = \mathbb{E}_{\boldsymbol{\mu}}\left[L(\boldsymbol{\mu}, \widehat{\boldsymbol{\mu}})\right].$$

Minimizing the risk function (mean squared error) yields the **James-Stein estimator**:

$$\widehat{\mu}_i^{JS} = \left(1 - \frac{(N-2)}{S^2}\right) X_i, \qquad \text{where} \qquad S^2 = \sum_{i=1}^{N} X_i^2$$

This estimator shrinks the naïve estimator toward 0 by a factor $((N-2)/S^2)$

The risk of this estimator is strictly smaller than the risk of X, for all $\boldsymbol{\mu}$ (when $N \geq 3$).

James-Stein Shrinkage Estimators

Efron-Morris estimator:

$$\mu_i^{EM} = \bar{X} + \left(1 - \frac{(N-3)}{S^2}\right)(X_i - \bar{X}) \qquad \text{where} \qquad S^2 = \sum_{i=1}^{N}(X_i - \bar{X})^2$$

Paradox: The X_i are i.i.d., so estimating each of the μ_i separately seems intuitive (least-squares estimator). Yet, combining the X_i provides a better estimate of the means.

Generalization: Optimal estimator of the mean should take the form, for $0 < \delta \leq 1$:

$$\widehat{\mu}_i^{JS} = \delta\mu_0 + (1-\delta)\widehat{\mu}_i \qquad \text{where} \qquad \widehat{\mu}_i = \frac{1}{T}\sum_{i=1}^{T} X_i$$

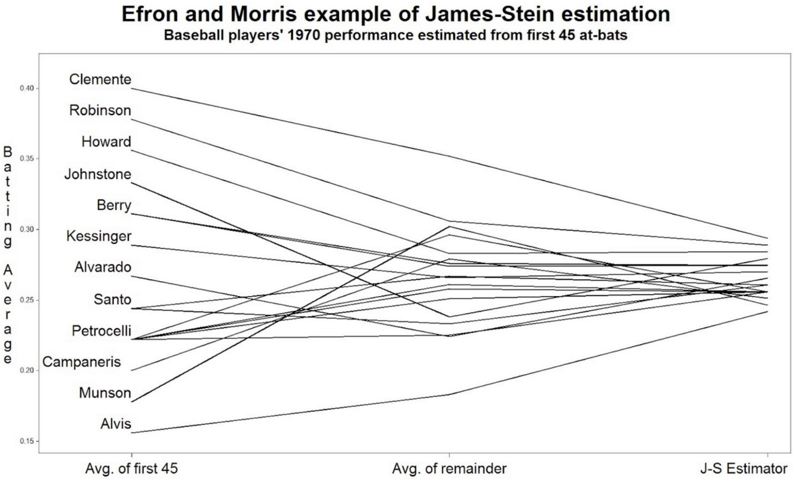
Sample means are "shrunk" by factor δ to fixed values $\boldsymbol{\mu}_0$.

Paradox: Any shrinkage target μ_0 results in less risk than sample means.

Common shrinkage targets: zero, grand (cross-sectional) mean, theoretical or sensible values

James-Stein Shrinkage Estimators: Example

Efron and Morris (1977): Is the batting average over the first 45 times at bat a good predictor for the remainder of the season? (grand average $= 0.265$)



James-Stein Shrinkage Estimators

Interpretation of the graph: The J-S estimator assumes that high-score players (Clemente) have been lucky so far and the low-score players (Alvis) have been unlucky. For the rest of the season, their scores should go back close to the grand mean.

Shrinkage estimator trades off higher bias for lower variance.

Optimal trade-off:

$$\delta^* = \min\left(1, \frac{(N-2)/T}{(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}_0)}\right)$$

Properties of δ^*:

- increases in number of unknowns N
- decreases in sample size T
- decreases in dispersion of sample means from shrinkage target

Different loss functions lead to different optimal shrinkage factors.

Objectives of the Lecture

- Constrained Portfolio Optimization
- Portfolio Resampling
- Factor Models
- Stein's Paradox
- **Shrinkage Estimator**

Shrinkage Estimator – Estimators of the Covariance Matrix

Estimators with decreasing structure, decreasing bias, increasing estimation error:

- **Single-factor model**
  - Relatively little estimation error
  - Tend to be misspecified and can be severely biased
- **Multifactor models**
  - Estimation error increases
  - Bias is reduced
- **Sample covariance matrix**
  - Subject to a lot of estimation error when N is large relative to T
  - Unbiased (expected value equals true covariance matrix)

Shrinkage Estimator

The shrinkage estimator finds a compromise between

- the sample covariance matrix ($\mathbf{S}$)
- and a highly structured estimator ($\mathbf{F}$)

$$\widehat{\mathbf{\Sigma}}_{shrink} = \delta \mathbf{F} + (1 - \delta)\mathbf{S}, \qquad \text{where} \qquad 0 \leq \delta \leq 1$$

It is called shrinkage as it "shrinks" the sample covariance matrix towards the structured estimator.

δ is the shrinkage constant (weight given to the structured estimator). Measuring δ is the only difficult step in shrinkage estimation.

Shrinkage estimators are based on the very logic of J-S estimators.

Shrinkage Estimator

Any shrinkage estimator has three ingredients:

1. **Estimator without structure**
   - Obvious given the context.
   - In this case: the sample covariance matrix.
2. **Estimator with a lot of structure (shrinkage target)**

   Should fulfill two requirements:
   - involve only a small number of free parameters (that is, a lot of structure)
   - reflect important characteristics of the unknown quantity to be estimated
3. **Shrinkage constant**
   - there exists an optimal shrinkage constant
   - it minimizes the distance between the shrinkage estimator and the true covariance matrix

Notations and Assumptions

Notations

- $\mathbf{\Sigma}$: the (true) covariance matrix, with entries σ_{ij}
- $\mathbf{S}$: the sample covariance matrix, with entries s_{ij}
- $\mathbf{\Phi}$: the (true) shrinkage target covariance matrix, with entries ϕ_{ij}
- $\mathbf{F}$: the sample shrinkage target covariance matrix, with entries f_{ij}
- $R_{i,t}$: the return on stock i during period t
- $\bar{r}$: the average sample correlation

Assumptions

- Stock returns are i.i.d. through time
- N is finite, T goes to infinity
- Stock returns have finite fourth moment (so that CLT applies to sample variances and covariances)

Note: stock returns are **NOT** assumed to be normally distributed!

Shrinkage Constant

Choose the objective according to which the shrinkage intensity (constant) δ is optimal.

Considering the Frobenius norm of the difference between the shrinkage estimator and the true covariance matrix gives the quadratic **Loss Function**:

$$L(\delta) = \|\delta \mathbf{F} + (1-\delta)\mathbf{S} - \mathbf{\Sigma}\|_F^2$$

The goal is to find the shrinkage constant δ that **minimizes** the expected value of this loss, that is, the **Risk**:

$$R(\delta) = \mathbb{E}[L(\delta)] = \mathbb{E}\left[\|\delta \mathbf{F} + (1-\delta)\mathbf{S} - \mathbf{\Sigma}\|_F^2\right]$$

Shrinkage Constant

The Frobenius norm of a (N, N) symmetric matrix with entries $z_{ij} i, j$ is defined by $\|Z\|_F^2 = \sum_{i=1}^N \sum_{j=1}^N z_{ij}^2$. This gives the Risk function:

$$
\begin{aligned}
R(\delta) = \mathbb{E}\left[L(\delta)\right] &= \sum_{i=1}^N \sum_{j=1}^N \mathbb{E}\left[\left(\delta f_{ij} + (1-\delta)s_{ij} - \sigma_{ij}\right)^2\right] \\
&= \sum_{i=1}^N \sum_{j=1}^N \mathrm{Var}\left[\delta f_{ij} + (1-\delta)s_{ij} - \sigma_{ij}\right] + \sum_{i=1}^N \sum_{j=1}^N \left(\mathbb{E}\left[\delta f_{ij} + (1-\delta)s_{ij} - \sigma_{ij}\right]\right)^2 \\
&= \sum_{i=1}^N \sum_{j=1}^N \left[\delta^2 \mathrm{Var}[f_{ij}] + (1-\delta)^2 \mathrm{Var}[s_{ij}] + 2\delta(1-\delta)\mathrm{Cov}[f_{ij}, s_{ij}]\right] \\
&+ \sum_{i=1}^N \sum_{j=1}^N \left(\mathbb{E}\left[\delta(f_{ij} - \sigma_{ij}) + (1-\delta)(s_{ij} - \sigma_{ij})\right]\right)^2 \\
&= \sum_{i=1}^N \sum_{j=1}^N \left[\delta^2 \mathrm{Var}[f_{ij}] + (1-\delta)^2 \mathrm{Var}[s_{ij}] + 2\delta(1-\delta)\mathrm{Cov}[f_{ij}, s_{ij}] + \delta^2(\phi_{ij} - \sigma_{ij})^2\right]
\end{aligned}
$$

Shrinkage Constant

Minimizing the Risk with respect to δ yields :

$$R'(\delta) = 2\sum_{i=1}^{N}\sum_{j=1}^{N}\left[\delta \mathrm{Var}[f_{ij}] - (1-\delta)\mathrm{Var}[s_{ij}] + (1-2\delta)\mathrm{Cov}[f_{ij}, s_{ij}] + \delta(\phi_{ij} - \sigma_{ij})^2\right]$$

and

$$R''(\delta) = 2\sum_{i=1}^{N}\sum_{j=1}^{N}\left[\mathrm{Var}[f_{ij} - s_{ij}] + (\phi_{ij} - \sigma_{ij})^2\right] > 0$$

Setting $R'(\delta)$ and solving for δ^* yields

$$\delta^* = \frac{\sum_{i=1}^{N}\sum_{j=1}^{N}(\mathrm{Var}[s_{ij}] - \mathrm{Cov}[f_{ij}, s_{ij}])}{\sum_{i=1}^{N}\sum_{j=1}^{N}(\mathrm{Var}[f_{ij} - s_{ij}] + (\phi_{ij} - \sigma_{ij})^2)}$$

Shrinkage Constant

Ledoit and Wolf (2003) show that δ^* behaves (asymptotically) like a constant κ/T.

The **shrinkage constant** δ^* can be written as

$$\delta^* = \frac{\kappa}{T} = \frac{1}{T}\frac{\pi - \rho}{\gamma}$$

- π: sum of asymptotic variances of the entries of the sample covariance matrix $\times\sqrt{T}$
- ρ: sum of asymptotic covariances of the entries of the shrinkage target with the entries of the sample covariance matrix $\times\sqrt{T}$
- γ: a measure of the misspecification of the (population) shrinkage target (relative to the true covariance matrix)

The shrinkage constant:

- increases in the error of the sample covariance matrix (through π)
- decreases in the misspecification of the shrinkage target model (through γ)
- ρ measures the covariance between the estimation errors of $\mathbf{F}$ and $\mathbf{S}$.

Shrinkage Constant

If κ were known, we could use κ/T as the shrinkage intensity. Ledoit and Wolf (2003) show that this optimal value behaves (asymptotically) like a constant.

- **Consistent estimator for π (estimator of $\mathrm{Var}[s_{ij}]$):**

$$\widehat{\pi} = \sum_{i=1}^{N}\sum_{j=1}^{N}\widehat{\pi}_{ij} \qquad \text{where} \qquad \widehat{\pi}_{ij} = \frac{1}{T}\sum_{t=1}^{T}[(R_{i,t}-\bar{R}_i)(R_{j,t}-\bar{R}_j)-s_{ij})]^2$$

- **Consistent estimator for ρ (estimator of $\mathrm{Cov}[f_{ij}, s_{ij}]$):**

$$\widehat{\rho} = \sum_{i=1}^{N}\widehat{\pi}_{ij} + \sum_{i=1}^{N}\sum_{j=1}^{N}\frac{\bar{r}}{2}\left(\sqrt{\frac{s_{jj}}{s_{ii}}}\widehat{\theta}_{ii,ij} + \sqrt{\frac{s_{ii}}{s_{jj}}}\widehat{\theta}_{jj,ij}\right)$$

where

- $\widehat{\theta}_{ii,ij} = \frac{1}{T}\sum_{t=1}^{T}[(R_{i,t}-\bar{R}_i)^2 - s_{ii}][(R_{i,t}-\bar{R}_i)(R_{j,t}-\bar{R}_j)-s_{ij})]$
- $\widehat{\theta}_{jj,ij} = \frac{1}{T}\sum_{t=1}^{T}[(R_{j,t}-\bar{R}_j)^2 - s_{jj}][(R_{i,t}-\bar{R}_i)(R_{j,t}-\bar{R}_j)-s_{ij})]$

- **Consistent estimator for γ (estimator of $\mathrm{Var}[f_{ij} - s_{ij}] + (\phi_{ij} - \sigma_{ij})^2$):**

$$\widehat{\gamma} = \sum_{i=1}^{N} \sum_{j=1}^{N} (f_{ij} - s_{ij})^2$$

All pieces put together yield a consistent estimator for $\delta^{\star}$:

$$\widehat{\delta} = \max\left(0, \min\left(1, \ \frac{\widehat{\kappa}}{T}\right)\right).$$

N.B.: $\widehat{\delta}$ is truncated at 0 and 1 (finite-sample possibility).

Shrinkage Target

Nowhere the model specifies what the shrinkage target is.

The formula for the shrinkage constant remains valid for any shrinkage target, as long as F

- is an asymptotically biased estimator of the covariance matrix
- satisfies a set of weak regularity conditions

Constant-Correlation Model (Ledoit and Wolf, 2004, JPM):

$$f_{ii} = s_{ii}, \qquad f_{ij} = \bar{r}\sqrt{s_{ii}s_{jj}} \quad (i \neq j)$$

where $\bar{r}$ is the average sample correlation.

Single-Index Covariance Matrix (Ledoit and Wolf, 2004, JEF):

$$\mathbf{F} = \beta\beta' \, \widehat{\sigma}_m^2 + \widehat{\mathbf{D}},$$

i.e., the market-model (single-index) implied covariance matrix.

Empirical Results

Performance of the covariance matrix estimator measured by the variance of the optimal portfolio AFTER it is formed (out-of-sample performance)

- Monthly returns on US stocks (1972–1994) (N between 909 and 1314 firms)
- Covariance matrix estimated from year $t-10$ to year t
- Minimum Variance portfolio held over year $t+1$
  - Global minimum variance portfolio (unconstrained)
  - Minimum variance portfolio, expected return $=$ 20% (constrained)
  - Short sales allowed in both cases
- Compute the out-of-sample standard deviation of the investment strategy over the 23-year period.

Empirical Results – Risk of minimum variance portfolios

Risk of minimum variance portfolios

	Standard deviation unconstrained	Standard deviation constrained
Identity	17.75 (0.44)	17.94 (0.42)
Constant correlation	14.27 (0.19)	16.30 (0.29)
Pseudo-inverse	12.37 (0.23)	13.73 (0.32)
Market model	12.00 (0.16)	13.77 (0.27)
Industry factors	10.84 (0.17)	12.32 (0.23)
Principal components	10.31 (0.16)	11.30 (0.22)
Shrinkage to identity	10.21 (0.17)	11.11 (0.21)
Shrinkage to market	9.55 (0.15)	10.43 (0.20)

"Unconstrained" refers to the global minimum variance portfolio, while "constrained" refers to the minimum variance portfolio with 20% expected return. Standard deviation is measured out-of-sample at the monthly frequency, annualized through multiplication by $\sqrt{12}$ and expressed in percents. Standard errors on these standard deviation estimates are reported in parenthesis.

Source: Ledoit and Wolf (JEF, 2003)

Empirical Results – Statistics on portfolio weights

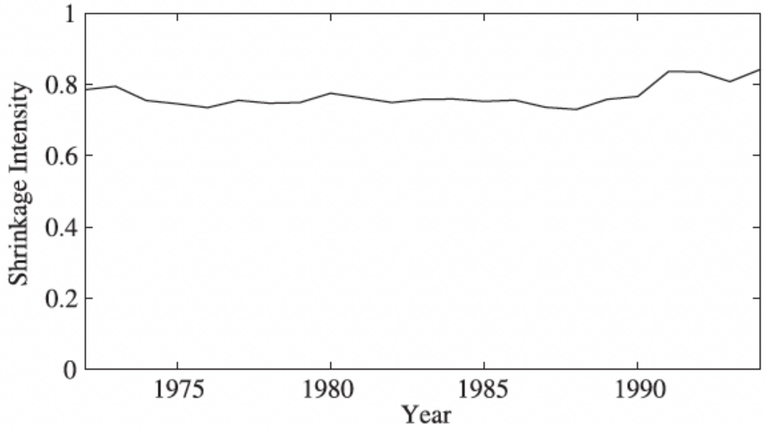
Weight descriptors

	Turnover	Short interest	Lowest weight	Highest weight	Cash position
Identity	6	0	0.09	0.09	2.50
Constant correlation	24	68	−0.17	2.86	−1.40
Pseudo-inverse	96	99	−1.16	1.12	1.46
Market model	23	51	−0.41	2.10	0.13
Industry factors	43	89	−1.08	2.81	0.54
Principal components	49	80	−0.84	2.95	0.78
Shrinkage to identity	71	113	−1.23	1.19	0.99
Shrinkage to market	61	98	−1.01	3.81	0.78

These are for the global minimum variance portfolio, expressed in percents, and averaged over the 23 years in our sample. A short interest of 68%, say, means that for every dollar invested in the portfolio we short 68 cents worth of stocks, while buying $1.68 worth of other stocks. Annual turnover above 100% is possible because of short sales.

Source: Ledoit and Wolf (JEF, 2003)

Empirical Results – Optimal shrinkage intensity estimate δ^*



Source: Ledoit and Wolf (JEF, 2003)

Conclusion

At one extreme: single-factor covariance matrix.

At the other extreme: sample covariance matrix (N-factor model with no residuals).

Intuition: the best model lies somewhere in between.

This intuition is usually expressed mostly through K-factor models (with $1 < K < N$).

Ledoit and Wolf (shrinkage estimators) recognize that:

- Single-index covariance matrix has a lot of bias (misspecified structural assumption), but little estimation error.
- Sample covariance matrix is unbiased (asymptotically), but has a lot of estimation error.

Conclusion

Fundamental principle of statistical decision theory: there exists an interior optimum in the trade-off between bias and estimation error.

Stein (1956): this optimal trade-off is attainable through a properly weighted average of the biased and the unbiased estimates.

This is called "shrinking" the unbiased estimator full of estimation error towards a target, which is the biased estimator.

It is an alternative to multi-factor models.

Main advantage: does not require knowledge of the number and nature of factors.

End of Lecture 4